DrBinati MedAssist

AI-Powered Multi-Modal Medical Analysis

Presentation | Cyprian Fusi

Presentation Agenda

1. **Problem Framing & AI Suitability** (2-3 min)
    - Business problem, user value, and why GenAI
    - Success metrics and key constraints
2. **Data & Model Approach** (4-5 min)
    - Data sources, processing pipeline, and model choices
3. **System Architecture & Integration** (4-5 min)
    - High-level architecture and component integration
4. **Key Trade-offs & Decision-Making** (3-4 min)
    - Technical decisions and alternatives considered
5. **Production Readiness & Operations** (3-4 min)
    - Monitoring, responsible AI, and evolution strategy

SECTION 1

Problem Framing & AI Suitability

The Healthcare Challenge

Physicians face information overload from multiple data sources:

- **ECG signals** with complex waveform patterns
- **Medical imaging** (X-rays, CT, MRI) requiring interpretation
- Unstructured **clinical notes** with critical patient history

Current challenges:

- **Time-consuming** manual correlation of data modalities
- **Risk of missing critical patterns** across data types
- **Specialist shortages** in underserved regions
- **Cognitive burden** during high-volume patient care

Why Generative AI?

GenAI advantages over traditional approaches:

- **Multi-modal understanding**: Process images + text + signals together
- **Contextual reasoning**: Correlate findings across modalities
- **Natural language output**: Human-readable interpretations

Why NOT 'rules/heuristics' alone:

- **Medical patterns too complex** for hard-coded rules
- **Need to handle novel combinations** of findings
- Requires **synthesis of diverse**, unstructured data

RAG enhancement:

- Retrieves **similar historical cases** for **context**
- **Grounds AI responses** in institutional knowledge

Success Metrics & Constraints

Success Metrics

Business Level:

- **Time saved** per patient analysis
- Clinician **satisfaction scores**
- Report **generation efficiency**

Model Level:

- **Analysis accuracy** vs expert review
- **Processing latency** < 60 seconds
- **Cost per analysis** < $0.05
- **RAG retrieval relevance** > 80%

Key Constraints

Data & Privacy:

- **HIPAA compliance** required
- **PHI** must be **de-identified**
- **Audit trail** for all analyses

Technical & Resource:

- **Budget**: $50/day, $1000/month
- **Single developer** (MSc project)
- **No access** to NVIDIA A100 GPUs
- PhysioNet data (**10TB**) not available

SECTION 2

Data & Model Approach

Multi-Modal Data Sources

ECG Signals:

- **Formats**: WFDB (PhysioNet), CSV, NumPy arrays
- **Features**: Heart rate, HRV metrics, R-peaks, intervals
- **Outputs**: Waveform visualizations + spectrograms

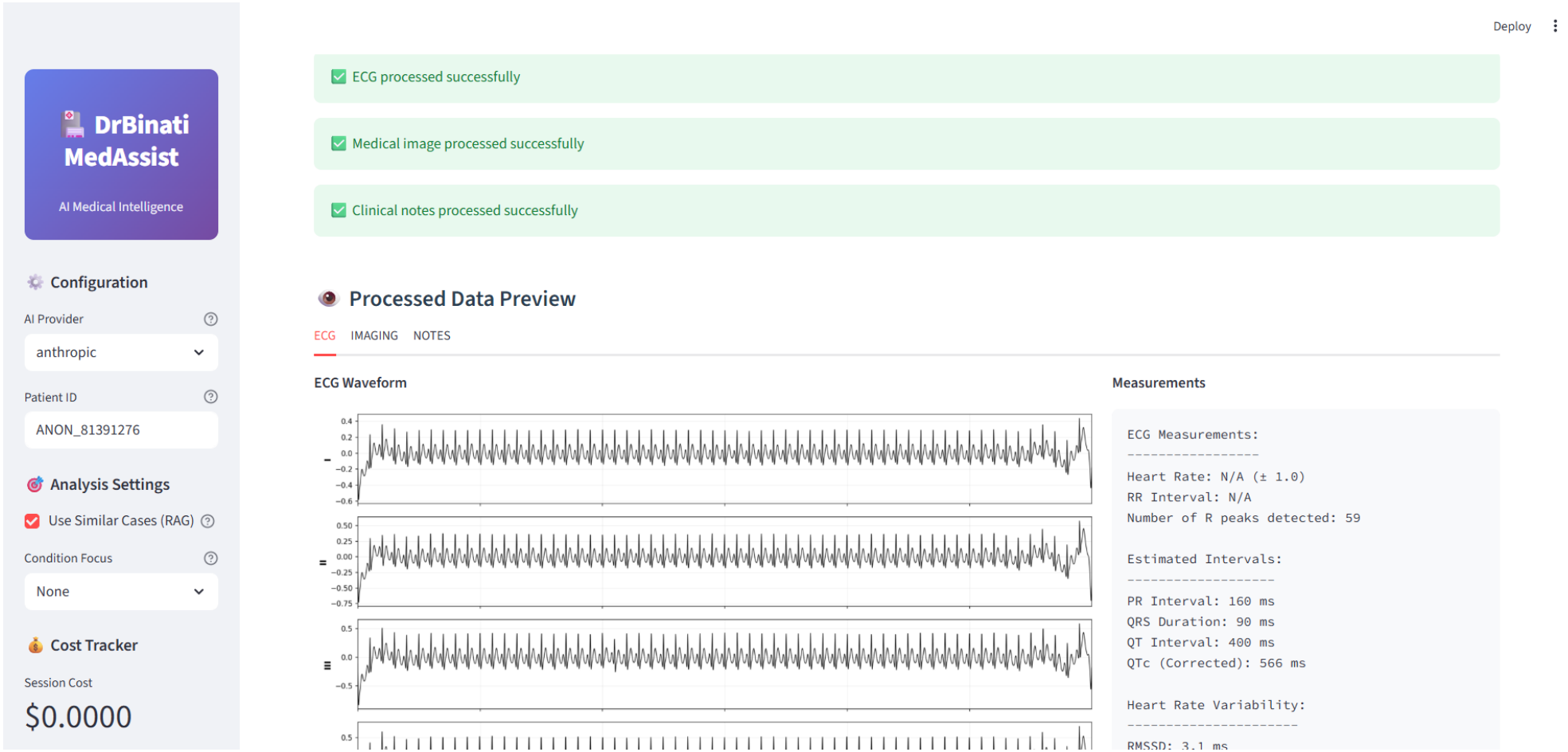
Medical Imaging:

- **Formats**: DICOM, NIfTI, PNG, JPG
- **Processing**: CLAHE enhancement, normalization
- DICOM **anonymization** for PHI removal

Clinical Notes:

- **De-identification** using Microsoft Presidio
- **Section extraction** (Chief Complaint, Assessment, Plan)
- Maximum **50,000 characters** per document

Data Processing Pipeline



ECG waveform processing with measurements extraction

Model Choice & Rationale

Primary Models

Claude 3.5/3.7 Sonnet (Anthropic):

- **Strong vision** capabilities
- Excellent medical **reasoning**
- **Cost-effective** for production
- Always updated and state-of-the-art (SOTA)

GPT-4o/GPT-4 Turbo (OpenAI):

- Alternative provider option
- Comparable vision performance
- Different pricing structure
- Always updated and state-of-the-art (SOTA)

Why Context Engineering?

vs. Fine-tuning:

- No access to **A100 GPUs**
- **Insufficient labeled** medical data
- Danger of **catastrophic forgetting**.
- **Faster iteration** cycles

vs. Training from Scratch:

- **Prohibitive** compute **costs**
- Foundation models already **excel**
- Focus on **prompts** + **RAG context**

RAG System Design

Vector Database:

- **PostgreSQL** + **pgvector extension**
- **Medical-specific embeddings** (BiomedNLP-PubMedBERT)
- 768/1536-dimensional vectors

Retrieval Strategy:

- Store embeddings of past analyses
- Query similar cases during new analysis
- Include relevant history as context

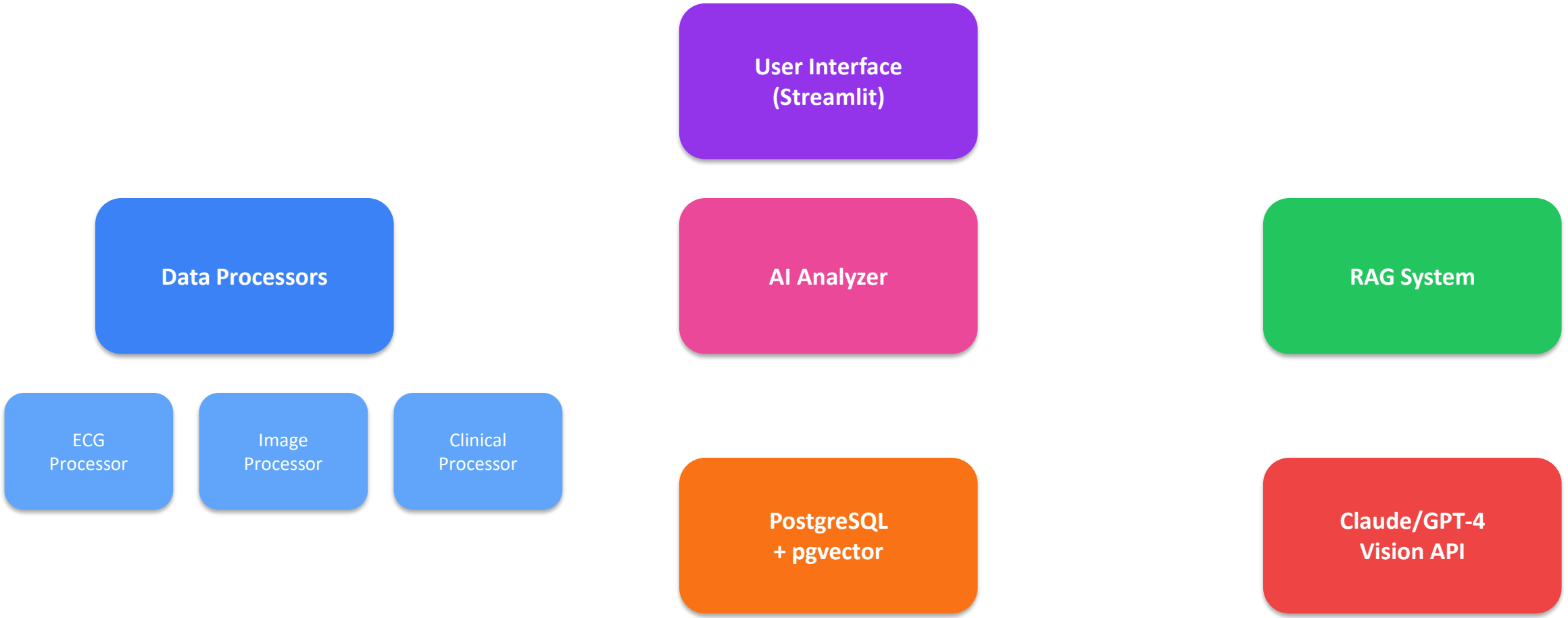
Benefits:

- Grounds responses in institutional data
- Enables comparative diagnosis support
- Maintains searchable case history

SECTION 3

System Architecture & Integration

System Architecture



Component Overview

User Interface Layer:

- Streamlit web application
- Three-column data upload (ECG, Imaging, Notes)
- Real-time cost tracking sidebar

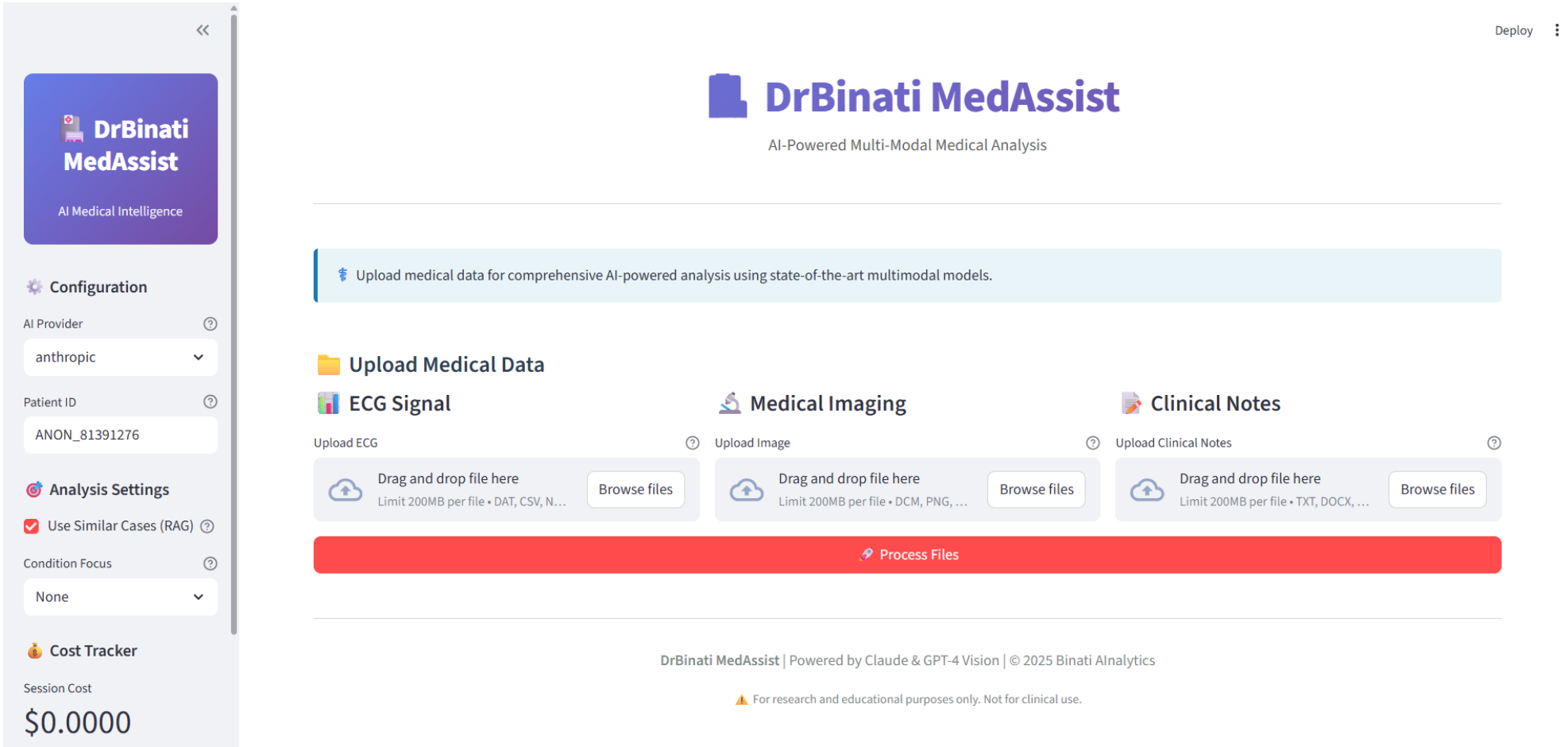
Processing Layer:

- Specialized processors for each modality
- Feature extraction and visualization generation
- PHI **de-identification** pipeline

AI Analysis Layer:

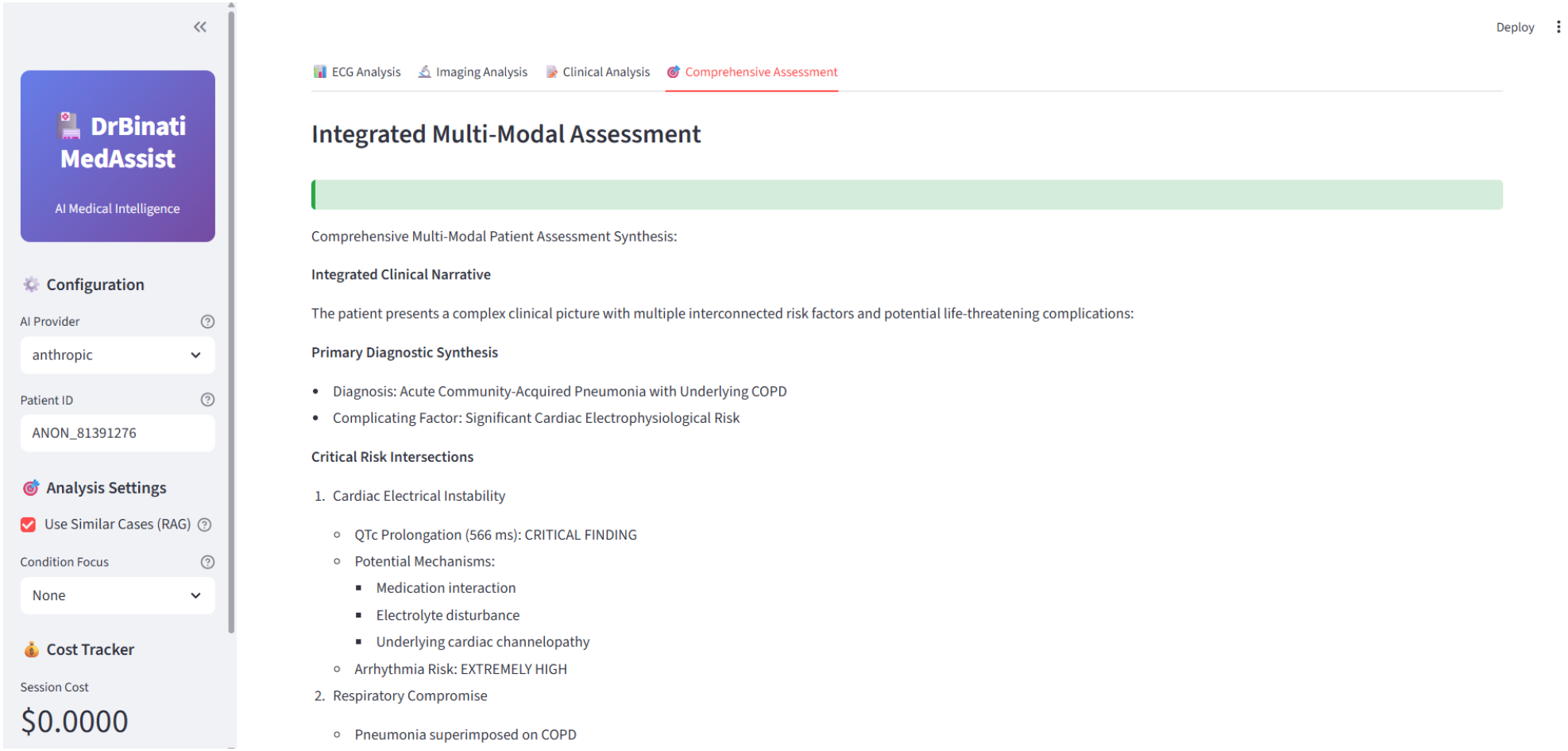
- MultiModalAnalyzer orchestration
- Condition-specific prompt selection
- Vision API integration (Claude/GPT-4)

User Interface - Main Dashboard



DrBinati MedAssist main interface with multi-modal upload

Analysis Results - Comprehensive Assessment



Integrated multi-modal patient assessment synthesis (partial display)

Integration & Ownership Boundaries

AI Component Responsibilities

Data Processing:

- Format conversion & validation
- Feature extraction
- PHI de-identification

Analysis:

- Multi-modal synthesis
- RAG context retrieval
- Report generation

Application Layer

User Interface:

- File upload management
- Configuration controls
- Results visualization

Infrastructure:

- Authentication (JWT)
- Rate limiting (60 req/min)
- Audit logging

SECTION 4

Key Trade-offs & Decision-Making

Technical Decisions

Decisions Made

API vs. Self-hosted Models:

- Chose: Claude/GPT-4 APIs
- Reason: No A100 GPU access

PostgreSQL vs. Specialized Vector DB:

- Chose: PostgreSQL + pgvector
- Reason: Simpler ops, HIPAA path

Streamlit vs. Custom Frontend:

- Chose: Streamlit
- Reason: Rapid prototyping, MSc scope

Alternatives Considered

Rejected: Training custom model

- PhysioNet: 10TB storage needed
- Data not openly accessible
- Compute costs prohibitive

Rejected: Fine-tuning LLaVA/Med-PaLM

- Requires significant GPU resources
- Medical fine-tuning data limited
- Potential of “Catastrophic Forgetting”
- Foundation/API models already performant

Accuracy vs. Latency vs. Cost Trade-offs (I)

Accuracy Optimization:

- Use Claude Sonnet for complex cases (higher accuracy)
- Include RAG context for institutional grounding
- Multi-pass analysis for comprehensive reports

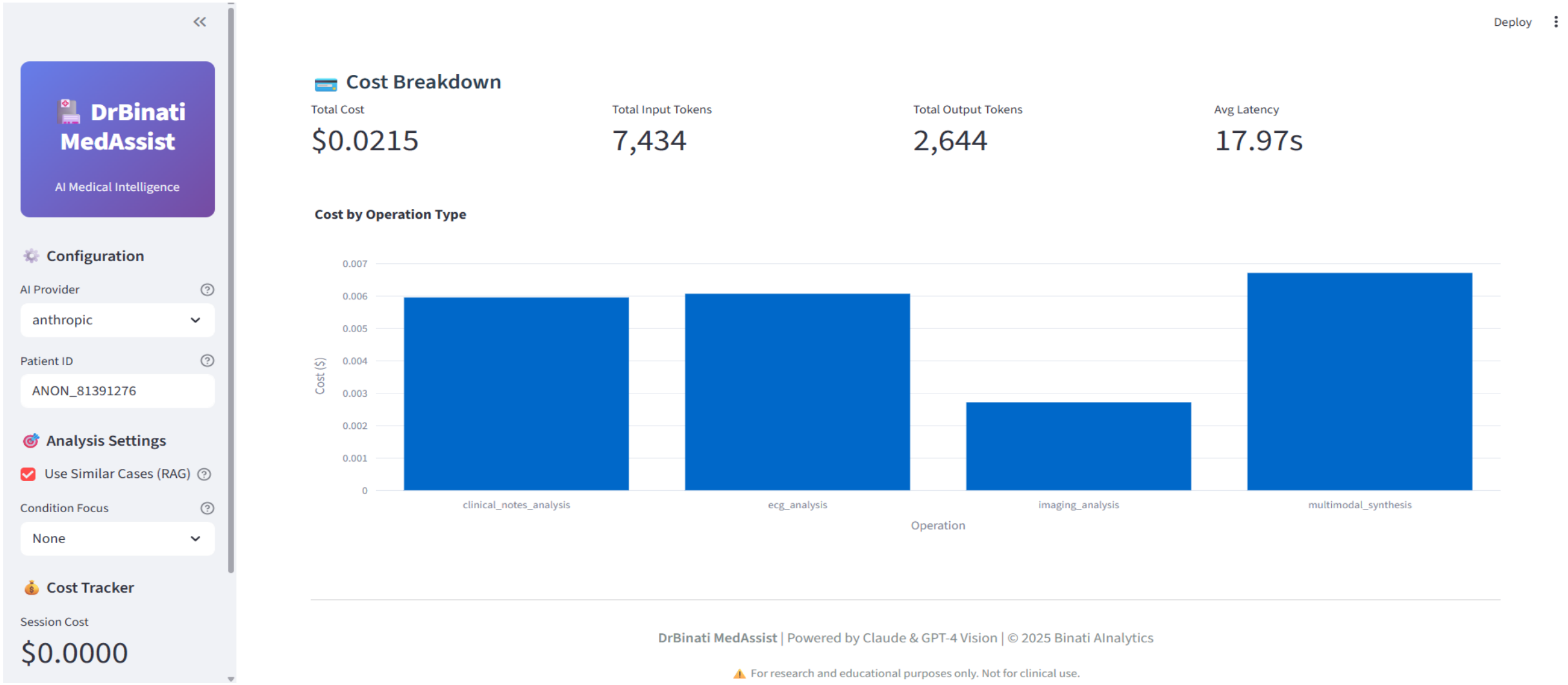
Latency Considerations:

- Target: < 60 seconds for full analysis
- Actual: 30-60 seconds with all modalities
- Parallel processing of independent modalities

Cost Management:

- Claude Haiku for simple queries (~$0.40/image)
- Daily budget alerts at 80% threshold
- Per-analysis cost tracking (~$0.04-0.05 typical)

Accuracy vs. Latency vs. Cost Trade-offs (II)



Latency and Cost Breakdown

Explaining to Non-AI Stakeholders

The 'Smart Assistant' Analogy:

- System is like having a specialist assistant
- Reviews all data sources from a patient simultaneously
- Provides structured findings / diagnoses for physician review

Key Points for Stakeholders:

- **NOT** a replacement for clinical judgment
- Decision support tool, not decision maker
- Always requires physician oversight

Managing Uncertainty:

- Confidence indicators in outputs
- Clear 'limitations' section in reports
- Iterative refinement based on feedback

SECTION 5

Production Readiness & Operations

Monitoring Strategy

Model Performance

Runtime Monitoring:

- Processing time per modality
- Token usage and costs
- API error rates

Quality Metrics:

- Physician feedback scores
- Report completeness checks
- Hallucination detection

Data & Infrastructure

Data Quality:

- Input validation checks
- Format compatibility logging
- De-identification success rates

System Health:

- Database connection pools
- API latency tracking
- Budget utilization alerts

Handling Drift & Retraining

Data Drift Detection:

- Monitor input data distributions
- Track embedding similarity patterns
- Alert on unusual input characteristics

Model Lifecycle:

- No traditional retraining (API-based models)
- Prompt refinement based on failure analysis
- RAG knowledge base updates

Failure Modes & Degradation:

- Graceful fallback when API unavailable
- Clear error messaging to users
- Manual review flagging for edge cases

Responsible AI Considerations

Bias Mitigation:

- Diverse test cases across demographics
- Monitor for disparate performance
- Regular bias audits planned

Explainability:

- Structured output with reasoning
- Source attribution in RAG responses
- Confidence indicators in findings

Guardrails & Safety:

- **Clear disclaimer:** 'Not for clinical use' (research)
- Physician oversight required for any decisions
- PHI protection at every layer

System Evolution Strategy

Near-term Enhancements:

- Additional imaging modalities (ultrasound, PET)
- Expanded condition-specific prompts
- Mobile-responsive interface

Medium-term Goals:

- Integration with EHR systems (FHIR)
- Multi-language support
- Collaborative analysis features

Long-term Vision:

- Clinical validation studies
- Regulatory pathway exploration (FDA/CE)
- Edge deployment for low-connectivity settings

CONCLUSION

Summary & Questions

Key Takeaways

- **Problem**: Multi-modal medical data analysis is time-intensive and ineffective
- **Solution**: AI-powered synthesis using Claude/GPT-4 Vision
- **Architecture**: Modular design with specialized processors
- **RAG Enhancement**: Historical case retrieval for context
- **Trade-offs:** Balanced accuracy, latency, and cost
- **Production Ready**: Monitoring, compliance, and evolution path
- **Responsible AI**: Explainability, bias awareness, guardrails

- **DrBinati MedAssist**: Demonstrating the potential of multi-modal GenAI in healthcare decision support

Thank You

Questions?

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